



Buzzer Control in PC/SC Mode

DUALi Inc.

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We have our development center in South Korea to provide technical support. For any technical assistance can contact our technical support team as below;

Tel: +82 31 213 0074

e-mail : duali@duali.com

INTRODUCTION

This document defines the USB and serial communication protocol between DUALi's DE-620R and a host computer (DE-620R support USB only, but some of DUALi's module support RS-232 with same protocol of DE-620R. So this document includes the RS-232 option).

The protocols in this document are all for chip or COS developers and not for normal card application developers.

DUALi's readers and modules supports ISO7816, ISO14443 type A/B, ISO18092, my-d™, Mifare®, DESFire®, FeliCa™, ISO15693 and I-CODE cards and high communication speed depending on the chip used in the reader.

DE-620 supports ISO7816, ISO14443 type A/B, ISO18092, my-d™, Mifare®, DESFire® and FeliCa™ cards. (**DE-ABCM, DE-ABM5**)

DE-620R supports ISO7816, ISO14443 type A/B, my-d™, Mifare®, DESFire®, ISO15693 and I-CODE cards. (**DE-ABM3**)

DE-600L supports ISO7816, ISO14443 type A/B, my-d™, Mifare®, DESFire® cards. (**DE-ABM2, DE-ABM4**)

This document is dedicated for DE-620R. So, if you use other reader products of DUALi, those readers are possible to return code UNKNOWN COMMAND ERROR(23, 0x17), it means your reader or module doesn't support this command.

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1. Buzzer Control

Reader will make beep sound when card is approached to reader in PC/SC mode. This sound can be deactivated or activated with special command.

■ Command frame in Vendor mode

LEN-H	LEN-L	CMD	Data[0]	Data[1]	Data[2]
4		0x1D	0x1B	0x04	0x02(Deactivate Buzzer) 0x00(Activate Buzzer)

■ Response frame for others

LEN-H	LEN-L	Resp
1		OK(0x00)

e.g.

Buzzer Beep when RF and Contact Card is Connected

Send: 1D 1B0400

Receive:00

Do not make buzzer sound when RF and Contact Card is Connected

Send: 1D 1B0402

Receive:00

■ Command frame in PC/SC mode

Load keys APDU format (12byte)					
CLASS	INS	P1	P2	LC	DATA IN
0xFE	0x1D	0xFE	0xFE	0x03	0x1B,0x04,0x02 (Deactivate) 0x1B,0x04,0x00 (Activate)

■ Response in PC/SC mode

Load keys response format (3byte)		
STATUS CODE	SW1	SW2
0x00	0x90	0x00